

Double-click (or enter) to edit

labsheet =03 computer vision Priyanshu Sharma Cu240250980 section=(B) (47)

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
import cv2

image_path = "/content/pexels-friededia-30649280.jpg"
img = cv2.imread(image_path)

if img is None:
    print("Image not found. Check the file path.")
else:
    # Convert BGR to RGB
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    print("Image loaded successfully!")
    print("Image size:", gray.shape)
    print("priyanshu sharma cu240250980")
```

```
Image loaded successfully!
Image size: (5464, 8192)
priyanshu sharma cu240250980
```

```
gaussian = cv2.GaussianBlur(gray, (5, 5), 0)
```

```
median = cv2.medianBlur(gray, 5)
```

```
average = cv2.blur(gray, (5, 5))
```

```
laplacian = cv2.Laplacian(gray, cv2.CV_64F)

# Convert to unsigned 8-bit
laplacian = cv2.convertScaleAbs(laplacian)
```

```
# 6. Sobel Edge Detection
# -----

# Horizontal edges
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

# Vertical edges
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Convert to 8-bit
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)

# Combined Sobel
sobel_combined = cv2.addWeighted(
    sobel_x, 0.5,
    sobel_y, 0.5,
    0
)
```

```
# 7. Display Original and Filtered Images
# -----

plt.figure(figsize=(15, 10))

plt.subplot(3, 3, 1)
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(3, 3, 2)
plt.imshow(gaussian, cmap="gray")
```

```
plt.title("Gaussian Filter")
plt.axis("off")

plt.subplot(3, 3, 3)
plt.imshow(median, cmap="gray")
plt.title("Median Filter")
plt.axis("off")

plt.subplot(3, 3, 4)
plt.imshow(average, cmap="gray")
plt.title("Average Filter")
plt.axis("off")

plt.subplot(3, 3, 5)
plt.imshow(laplacian, cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")

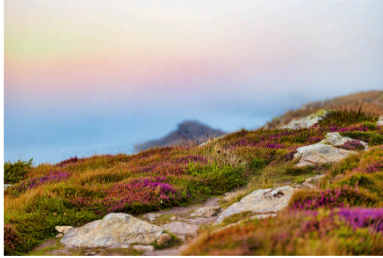
plt.subplot(3, 3, 6)
plt.imshow(sobel_x, cmap="gray")
plt.title("Sobel X - Horizontal")
plt.axis("off")

plt.subplot(3, 3, 7)
plt.imshow(sobel_y, cmap="gray")
plt.title("Sobel Y - Vertical")
plt.axis("off")

plt.subplot(3, 3, 8)
plt.imshow(sobel_combined, cmap="gray")
plt.title("Sobel Combined")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Original Image



Gaussian Filter



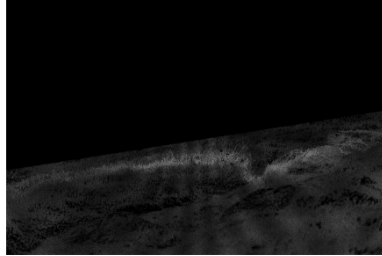
Median Filter



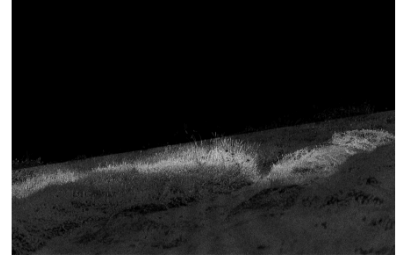
Average Filter



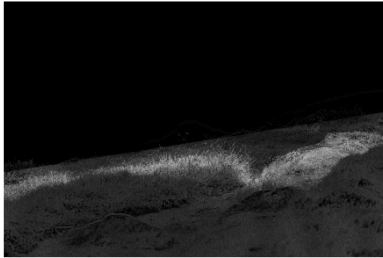
Laplacian Filter



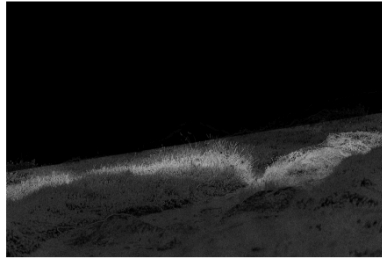
Sobel X - Horizontal



Sobel Y - Vertical



Sobel Combined



```
# 8. Save Output Images
# -----

cv2.imwrite("gaussian.jpg", gaussian)
cv2.imwrite("median.jpg", median)
cv2.imwrite("average.jpg", average)
cv2.imwrite("laplacian.jpg", laplacian)
cv2.imwrite("sobel_x.jpg", sobel_x)
cv2.imwrite("sobel_y.jpg", sobel_y)
cv2.imwrite("sobel_combined.jpg", sobel_combined)

print("\nOutput images saved successfully!")
```

Output images saved successfully!

```
# 9. Display Observations
# -----

print("\n===== OBSERVATIONS =====")

print("""
1. Gaussian Filter:
  - Smooths the image.
  - Reduces random noise.
  - Preserves important structures better than average filtering.
```

- ```
"""
```
2. Median Filter:
    - Very effective for salt-and-pepper noise.
    - Preserves edges better than average filtering.
  3. Average Filter:
    - Smooths the image.
    - Reduces noise.
    - Can blur edges and fine details.
  4. Laplacian Filter:
    - Detects rapid changes in intensity.
    - Enhances edges and fine details.
    - It is a high-pass filtering technique.
  5. Sobel Filter:
    - Detects image edges.
    - Sobel X detects changes in the horizontal direction.
    - Sobel Y detects changes in the vertical direction.
    - Combined Sobel shows edges in both directions.
- ```
"""
```

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 - Preserves important structures better than average filtering.
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